

Tropical Cyclone Structure

MEA 458

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Material source (text and figures): Roger Smith's notes and other assorted sources.

By the end of the lecture, you should be able to:

- ▶ identify the eye, eyewall, stratiform region, principal rainband, secondary bands, and outer convection;
- ▶ sketch the primary and secondary circulations in radius–height coordinates;
- ▶ describe typical radial and vertical structures of tangential and radial wind;
- ▶ explain the warm/dry eye, moist eyewall, inversion, and warm-core/warm-ring distinction;

In general, we can classify atmospheric phenomena as wave or vortices

- ▶ Examples of atmospheric waves: Rossby, gravity, Kelvin waves
- ▶ A generic wave is described by its parameters (frequency, wavenumbers) and how these parameters are related to each other (a dispersion relationship).
- ▶ A wave does not transport fluid (parcels only oscillate about some equilibrium position).
- ▶ A vortex can be thought of as a localized swirling disturbance that carries fluid with it.

Two circulations: primary and secondary

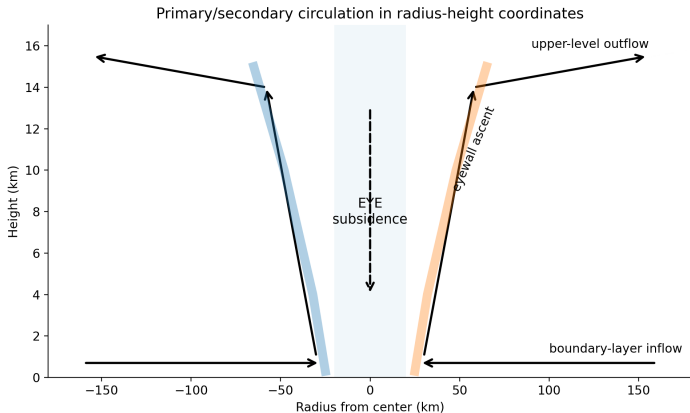
Primary circulation

- ▶ Dominantly azimuthal/tangential wind v .
- ▶ Near-gradient balance above the boundary layer.
- ▶ Maximum wind near the eyewall / radius of maximum wind (RMW).

Secondary circulation

- ▶ Frictional low-level radial inflow $u < 0$.
- ▶ Deep ascent concentrated in the eyewall.
- ▶ Upper-tropospheric outflow.
- ▶ Subsidence within the eye and parts of the outer core.

Radius–height view of the circulation

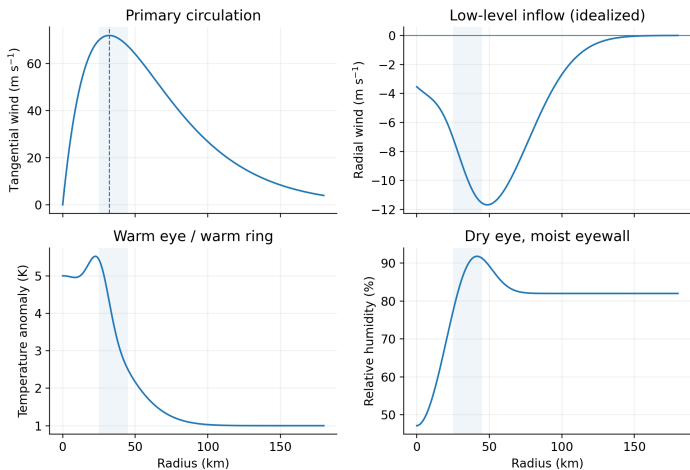


Tangential wind: the dominant component

- ▶ Tangential Wind (v) increases rapidly outward through the eye and peaks near the Radius of maximum wind (RMW).
- ▶ Outside the RMW, v decreases more gradually with radius; the outer circulation can extend hundreds of km.
- ▶ The maximum tangential wind generally decreases and shifts outward with height.

Representative radial profiles

Conceptual radial profiles synthesized from aircraft literature



Radial wind and the boundary layer

- ▶ Surface stress destroys angular momentum, producing cross-isobaric inflow.
- ▶ Inflow is strongest in the lowest 1–2 km and can persist somewhat higher in intense storms.
- ▶ Near the eyewall, the boundary layer can become supergradient; rapid deceleration and convergence favor strong ascent near/inside the RMW.
- ▶ Above the inflow layer, weak radial outflow or alternating inflow/outflow can occur depending on height and storm asymmetry.

Vertical motion: concentrated, asymmetric, intermittent

- ▶ The azimuthal-mean eyewall is ascending, but individual convective updrafts are much stronger than the mean.
- ▶ Strongest updraft can typically be found in the inner edge of the eyewall.
- ▶ Downdrafts occur within/adjacent to convection and in stratiform precipitation regions.

- ▶ Surface pressure is lowest in the eye; winds approach zero only near the circulation axis.
- ▶ The free-tropospheric eye is usually warmer and drier than the eyewall/environment because of subsidence.
- ▶ A low-level moist layer may support stratocumulus or a central “hub cloud.”
- ▶ A sharp inversion often separates moist lower-eye air from dry subsiding air above.

Hurricane Structure: Examples



Figure: Hurricane Florence (2018). source: NASA

Observed Structure of Tropical Cyclones

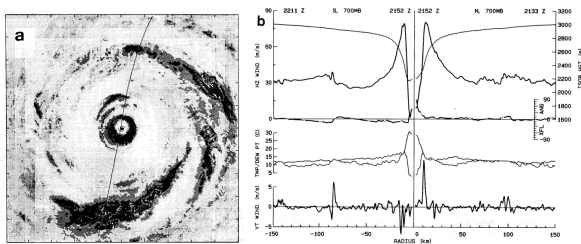


Figure: Observations from Hurricane Gilbert, 1988

(a) Plan-position indicator (PPI) radar reflectivity composite of Hurricane Gilbert at about 2200 UTC on 13 September 1988, when it was at maximum intensity near 19.9N, 83.5W. (b) Flight-level measurements from research aircraft. The abscissa is distance along a north-south pass through the centre. The top panel shows wind speed (dark solid line), 700 mb height (light solid line), and crossing angle ($\tan^{-1}u/v$, dash-dotted line). Winds are relative to the moving vortex centre. The middle panel shows temperature (upper curve) and dewpoint. When $TD > T$, both are set to $1/2(T + TD)$. The bottom panel shows vertical wind. [From (Black and Willoughby (1992))] source: Roger Smith's notes on tropical cyclones

Key features seen in Fig. 2a:

- 1 This radar composite image shows a tiny eye. The inner edge of the eyewall is just around 8km from the center, and the outer edge is less than 20km from the center. The eye is clear of deep clouds. Just outside the outer edge of the eyewall is another nearly clear region called the moat. As the aircraft flew across the moat at 3 km altitude, it was in rain beneath an overcast sky, and low stratocumulus obscured the surface.
- 2 Beyond the outer edge of the moat (75 km from the centre), the radar image shows precipitation organized into spirals that appear to be coalescing into a second ring of convection around the inner eye. This is a secondary eyewall that formed and contracted inwards.

Observed Structure of Tropical Cyclones

Key features seen in Fig. 2b:

- 1 The strongest horizontal wind (> 80 m/s) is in the eyewall, only 12 km from the calm at the axis of rotation. This is typical of a tropical cyclone, although in weaker storms the radius maximum wind speed is larger, ranging up to 50 km or more. Outside the eyewall, the wind drops abruptly to about 30 m/s at the outer edge of the moat and then rises to 35 m/s in the partial band of convection surrounding the moat.
- 2 The air temperature shows a steady rise as the aircraft flies inwards towards the eyewall and then a rapid rise as it enters the eye. **Thus the warmest temperatures are found in the eye itself, not in the eyewall clouds where the latent heat occurs.**
- 3 At most radii, the dewpoint depression is on the order of 4°C . The air is saturated only where convective vertical motions pass through flight level. Inside the eye, the temperature is greater than 28°C and the dewpoint is less than 0°C . These warm and dry conditions are typical of the eyes of extremely intense tropical cyclones.

Observed Structure of Tropical Cyclones

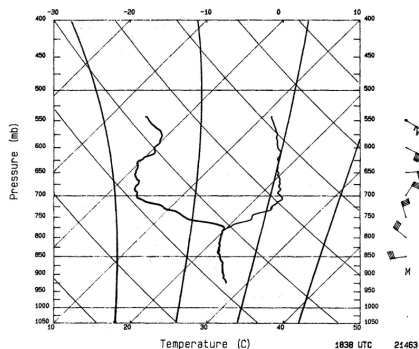


Figure: A dropsonde observation in the eye of Hurricane Hugo, near 14.7°N 54.8°W at 1838 UTC September 1989. Temperature is the right hand curve and dewpoint is the left. Nearly vertical curving lines are moist adiabats. Lines sloping up to the left are dry adiabats; those sloping up to the right are isotherms; and horizontal lines are isobars. [From Hawkins and Rubsam (1968) via Roger Smith]

Key features seen in Fig. 3:

- 1 Very dry and warm air above 700 hPa is indicative of subsidence inside the eye
- 2 An inversion at 700 mb separates very dry air with a dewpoint depression of about 20 °C from saturated air that follows a moist adiabat down to the sea surface.
- 3 Above the inversion, the subsiding air flows downward as part of a thermally indirect, forced subsidence in the eye. It is moistened a little by entrainment from the eyewall and evaporation of virga.
- 4 Below the inversion, the air is cooler and nearly saturated as a result of inflow under the eyewall, inward mixing, and evaporation from the sea inside the eye.

Observed Structure of Tropical Cyclones

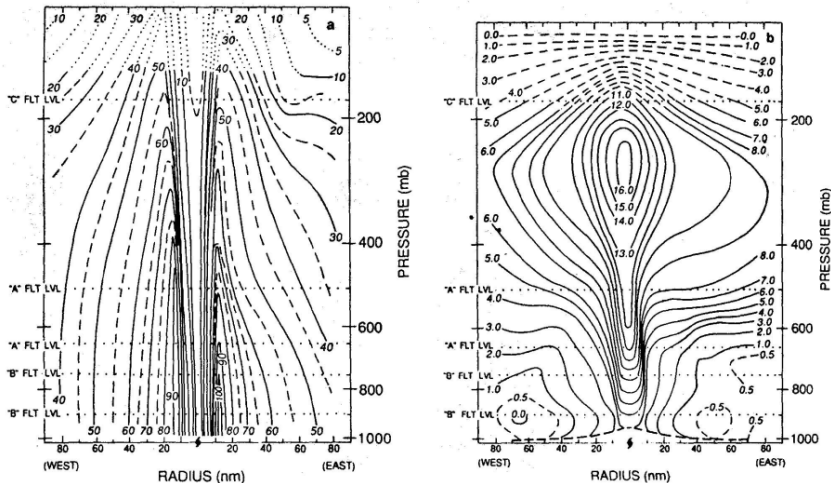


Figure: Vertical cross-sections of (left) azimuthal wind (kt), and (right) temperature anomaly (K) in Hurricane Hilda of 1964. [From Hawkins and Rubsam (1968)]

Observed Structure of Tropical Cyclones

Key features seen in Fig. 4:

- 1 The primary circulation (this is the swirling, tangential flow) in Hurricane Hilda is strongest just above the frictional boundary layer. Below 500 mb, it has little vertical shear, but in the upper troposphere, it becomes weaker and less symmetric, and the radial outflow is a large fraction of the swirling motion. Near the tropopause beyond 200 km radius, the vortex turns anticyclonic because of angular momentum loss to the sea on the inflow leg of the secondary circulation (Riehl, 1968).
- 2 The anomalous temperature cross-section (anomaly here is defined relative to the mean temperature at each level) shows that the hurricane is warm core. It also shows that the eye of the hurricane is warmest at the upper-levels (400 to 200 hPa) relative to the environment. This indicates subsidence at upper-levels.

Observed Structure of Tropical Cyclones

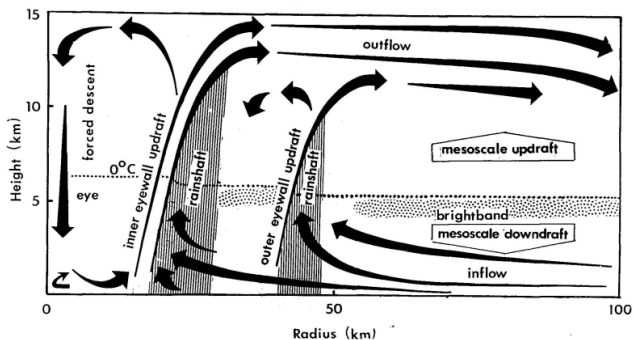


Figure: Schematic of the secondary circulation and precipitation distribution for a mature tropical cyclone [From Willoughby (1995)]

Observed Structure of Tropical Cyclones

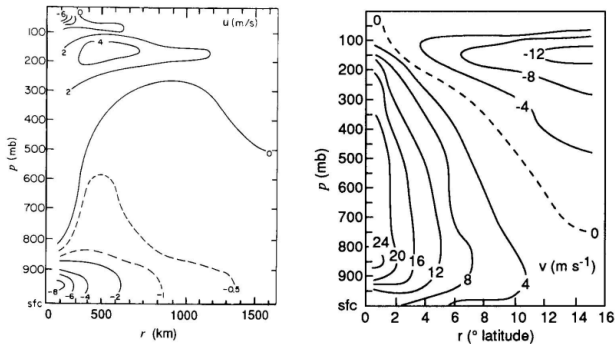


Figure: Vertical cross section of (a) the mean radial wind (m/s), (b) the mean tangential wind (m/s) Analysis is a composite of data collected in many storms. [From Gray (1979), Frank (1977)]

Key features seen in Fig. 7:

- ➊ Radial flow is predominantly in the lower and upper layers. It is inward at low-levels and outward at upper levels
- ➋ Tangential flow is cyclonic in a deep layer and anticyclonic in the upper-layer
- ➌ Both radial inflow and cyclonic tangential flow increase towards the center

Observed Structure of Tropical Cyclones

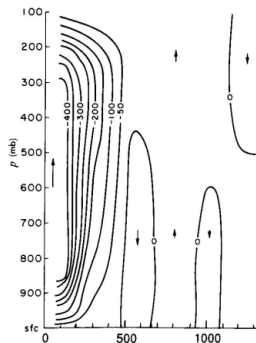


Figure: Vertical cross section of the mean vertical air motion (mb per day) in typhoons. Analysis is a composite of data collected in many storms. [From Gray (1979), Frank (1977)]

For more reading:

Roger Smith's TC notes

Hurricanes, by F. D. Marks. Encyclopedia of Atmospheric Sciences
(Second Edition) [See moodle for a pdf]